

Nuclear Physics Angular Momentum Coupling Guide

This document provides the theoretical background and practical usage guide for the `angular_momentum_coupling.py` utility. This tool calculates allowed final nuclear states based on quantum mechanical selection rules.

1. Theoretical Basis

The code implements the **Channel Spin Coupling Scheme**, which is a standard method for determining angular momentum conservation in nuclear reactions.

Notation

Symbol	Meaning
J_{target}	Spin of target nucleus (Python target input)
π_{target}	Parity of target nucleus (Python target input)
s_{particle}	Spin of transferred particle (Python particle input)
π_{particle}	Parity of transferred particle (Python particle input)
T_{particle}	Isospin of transferred particle
ℓ	Relative orbital angular momentum between the target and the particle
SS	Channel Spin = $J_{\text{target}} + s_{\text{particle}}$
J_{final}	Spin of the populated final nuclear state
π_{final}	Parity of the populated final nuclear state

Notation discipline: Use lowercase s_{particle} for the intrinsic spin of the transferred object (single nucleon, nucleon pair, or cluster). Uppercase SS is reserved for the channel spin formed by coupling J_{target} with s_{particle} . Do not use SS to describe the transferred object itself.

Conservation Laws

In any nuclear reaction $\text{Target} + \text{Particle} \rightarrow \text{Final_State}$, the following quantities are conserved:

1. **Total Angular Momentum** ($\vec{J}$):
$$\vec{J}_{\text{final}} = \vec{J}_{\text{target}} + \vec{s}_{\text{particle}} + \vec{\ell}$$
2. **Parity** (π):
$$\pi_{\text{final}} = \pi_{\text{target}} \times \pi_{\text{particle}} \times (-1)^{\ell}$$

Coupling Scheme Used in Code

The code performs the vector addition in two steps (Channel Spin representation):

1. **Calculate Channel Spin** ($\vec{S}$): Couples the target spin and particle spin.

$$\vec{S} = \vec{J}_{\text{target}} + \vec{s}_{\text{particle}}$$

Possible values: $|J_{\text{target}} - s_{\text{particle}}| \leq S \leq J_{\text{target}} + s_{\text{particle}}$

2. **Calculate Final Spin** ($\vec{J}_{\text{final}}$): Couples the channel spin with the orbital angular momentum.

$$\vec{J}_{\text{final}} = \vec{\ell} + \vec{S}$$

Possible values: $|\ell - S| \leq J_{\text{final}} \leq \ell + S$

Important: The "Particle" input to this tool represents the **intrinsic properties of the transferred particle**: spin s_{particle} , parity π_{particle} , and isospin T_{particle} .

2. Application to Different Reaction Types

To use the code correctly, you must identify the correct "Particle" input based on the reaction mechanism.

A. Resonant Capture Reactions

Examples: (p, γ) , (n, γ)

In these reactions, a projectile fuses with the target to form a compound nucleus (resonance).

- **Particle Input:** 1/2+

B. Single-Nucleon Transfer Reactions

Examples: (d, p) , (p, d) , $(^3\text{He}, d)$, $(d, ^3\text{He})$, (n, d) , (d, n) , (d, t) , (t, d) , (α, t) , (t, α) , $(^3\text{He}, \alpha)$, $(\alpha, ^3\text{He})$

In these reactions, a single nucleon is transferred between the projectile and the target. You need to input the properties of the **transferred nucleon**.

- **Particle Input:** 1/2+

C. Two-Nucleon Transfer Reactions

In these reactions, a pair of nucleons is transferred. The total spin of this pair depends on the coupling of the two nucleons.

1. Identical Nucleon Transfer (2n or 2p)

Reactions: (p, t) , (t, p) (2n transfer); $(^3\text{He}, n)$ (2p transfer)

For the transfer of two identical nucleons (2n or 2p) in the same shell model orbit, the **Pauli Exclusion Principle** dictates that their total wavefunction must be antisymmetric.

- If they are in a relative s-state (spatially symmetric, $L=0$), their **spin wavefunction must be antisymmetric**.
- An antiparallel spin state for two fermions corresponds to **$s_{\text{particle}}=0$** (spin singlet for the transferred pair).

- **NDS Policy:** For (p, t) , (t, p) , and $(^3\text{He}, n)$ reactions, it is standard to assume the transferred pair is in a relative ss state (**$s_{\text{particle}}=0$**).
- **Particle Input:** 0+ for transferring $s_{\text{particle}}=0$, $T_{\text{particle}}=1$ particle pair

2. Neutron-Proton Transfer (1n1p)

Reactions: (α, d) , (d, α) , $(^3\text{He}, p)$, $(p, ^3\text{He})$

For the transfer of a neutron-proton pair, the selection rules depend on the projectile/ejectile properties.

- **(α, d) and (d, α) :**
 - Both d and α have $T=0$. Thus, only **$T=0$** transfer is allowed.
 - Antisymmetry requires $s_{\text{particle}}+T$ to be odd (for $L=0$). Since $T=0$, the transferred pair must be spin triplet, i.e., **$s_{\text{particle}}=1$** .
 - **Particle Input:** 1+ for transferring $s_{\text{particle}}=1$, $T_{\text{particle}}=0$ particle pair
- **$(^3\text{He}, p)$ and $(p, ^3\text{He})$:**
 - Both particles have $s_{\text{particle}}=1/2$, $T_{\text{particle}}=1/2$.
 - Allowed transferred-pair components: **$(s_{\text{particle}}=0, T_{\text{particle}}=1)$** and **$(s_{\text{particle}}=1, T_{\text{particle}}=0)$** .
 - **Particle Input:**
 - 0+ for transferring $s_{\text{particle}}=0$, $T_{\text{particle}}=1$ particle pair
 - 1+ for transferring $s_{\text{particle}}=1$, $T_{\text{particle}}=0$ particle pair
 - Note: Both components can contribute.

D. Cluster Transfer Reactions

Examples: $(^6\text{Li}, d)$, $(^7\text{Li}, t)$

An alpha particle is transferred.

- **Particle Input:** 0+

E. Inelastic Scattering

Examples: (α, α') , (p, p')

- **(α, α') :** The alpha particle has spin 0. Since the projectile spin cannot flip, the angular momentum transfer is purely orbital ($\vec{\ell}$). This selectively excites **Natural Parity** resonances ($\pi = (-1)^\ell$) in the compound nucleus.
- **Particle Input:** 0+
- **(p, p') :** The proton ($s=1/2$) can undergo spin-flip. It allows both Isoscalar ($T=0$) and Isovector ($T=1$) transitions.
 - **Particle Input:** 0+ for non spin-flip
 - **Particle Input:** 1+ for spin-flip

F. Charge Exchange Reactions

Examples: (p, n) , $(^3\text{He}, t)$, $(^6\text{Li}, ^6\text{He})$

These reactions exchange a nucleon type ($n \leftrightarrow p$), changing the isospin projection by $\Delta T_z = \pm 1$. The total isospin change ΔT depends on the transition type:

- **Fermi Transitions** ($\Delta S=0, \Delta L=0, \Delta T=0, \Delta T_z=\pm 1$): Pure isospin operator without spin change. Excites the Isobaric Analog State (IAS) with $\Delta J^\pi = 0^+ \rightarrow 0^+$.
- **Gamow-Teller Transitions** ($\Delta S=1, \Delta L=0, \Delta T=0$ or $1, \Delta T_z=\pm 1$): Spin-isospin coupled operator. Excites states with $\Delta J = 0, 1$ (but not $0^+ \rightarrow 0^+$). Strong transitions from 0^+ targets populate 1^+ states.
- **$(^6\text{Li}, ^6\text{He})$** : Pure Gamow-Teller probe due to the $1^+ \rightarrow 0^+$ projectile transition. Fermi transitions are forbidden.
 - **Particle Input:** 1^+
- **(p, n) and $(^3\text{He}, t)$** : Mixed probes.
 - **Particle Input:** 0^+
 - **Particle Input:** 1^+

3. Interpreting the Output

When you run the code:

```
python .github/scripts/angular_momentum_coupling.py 3/2- 1/2+
```

You receive output like:

```
Target: J=3/2 π=- | Particle: s=1/2 π=+
Channel Spins S: 1, 2: from |3/2 - 1/2| to 3/2 + 1/2
-----
Wave   L     S     Final Jπ
-----
s       0     1     1-
s       0     2     2-
-----
p       1     1     0+, 1+, 2+
p       1     2     1+, 2+, 3+
-----
d       2     1     1-, 2-, 3-
d       2     2     0-, 1-, 2-, 3-, 4-
-----
```

- **Wave:** The orbital angular momentum (ℓ) between target and particle, expressed as spectroscopic notation (s, p, d, f, g, h, i for $\ell = 0, 1, 2, 3, 4, 5, 6$).
- **L:** Numeric value of orbital angular momentum (ℓ).

- **S:** Channel Spin value ($S = \vec{J}_{\text{target}} + \vec{s}_{\text{particle}}$). For $3/2^-$ target and $1/2^+$ particle: S ranges from $|3/2 - 1/2| = 1$ to $3/2 + 1/2 = 2$.
- **Final $J\pi$:** All allowed final spin-parity combinations for each (L, S) pair. Values are calculated from $|J_{\text{ell}} - S| \leq J_{\text{final}} \leq J_{\text{ell}} + S$. Parity is determined by $\pi_{\text{final}} = \pi_{\text{target}} \times \pi_{\text{particle}} \times (-1)^{\text{ell}}$.

Example Analysis: $^{59}\text{Cu}(p, \gamma)^{60}\text{Zn}$

- **Target:** ^{59}Cu ($3/2^-$)
- **Particle:** Proton ($1/2^+$)
- **Channel Spins:** $S = 1, 2$ (from $|3/2 - 1/2|$ to $3/2 + 1/2$)
- **Results:**
 - **s-wave capture** ($L=0$): Parity = $(-) \times (+) \times (-1)^0 = -$
 - $S=1$: $J^\pi = 1^-$
 - $S=2$: $J^\pi = 2^-$
 - **p-wave capture** ($L=1$): Parity = $(-) \times (+) \times (-1)^1 = +$
 - $S=1$: $J^\pi = 0^+, 1^+, 2^+$ (from $|1-1|$ to $1+1$)
 - $S=2$: $J^\pi = 1^+, 2^+, 3^+$ (from $|2-1|$ to $2+1$)
 - **d-wave capture** ($L=2$): Parity = $-$
 - $S=1$: $J^\pi = 1^-, 2^-, 3^-$
 - $S=2$: $J^\pi = 0^-, 1^-, 2^-, 3^-, 4^-$

4. Isospin Coupling

For transfer reactions, isospin (T) provides additional selection rules.

- **Single Nucleon Transfer:** Transferring one nucleon (n or p) has $T_{\text{particle}} = 1/2$.
- **Identical Nucleon Pair ($2n/2p$):** In a relative $L=0$ state, the Pauli Principle requires $s_{\text{particle}}=0$ (spin singlet for the transferred pair) and $T_{\text{particle}}=1$ (isospin triplet).
- **Neutron-Proton Pair (np):**
 - Deuteron-like ($s_{\text{particle}}=1, T_{\text{particle}}=0$)
 - Singlet ($s_{\text{particle}}=0, T_{\text{particle}}=1$)

Selection Rule: The final isospin is determined by vector addition:

$$\vec{T}_{\text{final}} = \vec{T}_{\text{target}} + \vec{T}_{\text{particle}}$$

Note: This tool calculates angular momentum (J^π) only. Isospin selection rules must be applied separately.

5. Summary of Transferred Particle Properties

Reaction Mechanism	Common Examples	Physical Process	s_{particle}	T_{particle}
Radiative Capture	$(p, \gamma), (n, \gamma)$	Projectile capture	$1/2$	$1/2$
	(α, γ)	Projectile capture	0	0

Reaction Mechanism	Common Examples	Physical Process	S_{particle}	T_{particle}
Inelastic Scattering	(α, α')	Excitation (Natural Parity)	0	0
	(p, p')	Excitation (non-flip / spin-flip)	0 and 1	0 and 1
One Neutron Transfer	$(d, p), (p, d), (t, d), (d, t), (\alpha, ^3\text{He}), (^3\text{He}, \alpha)$	Transfer of n	1/2	1/2
One Proton Transfer	$(d, n), (n, d), (^3\text{He}, d), (d, ^3\text{He}), (\alpha, t), (t, \alpha)$	Transfer of p	1/2	1/2
Two-Nucleon Transfer	$(p, t), (t, p), (^3\text{He}, n)$	Transfer of $2n$ or $2p$ ($L_{\text{rel}}=0$)	0	1
	$(\alpha, d), (d, \alpha)$	Transfer of np (Deuteron-like)	1	0
	$(^3\text{He}, p), (p, ^3\text{He})$	Transfer of np (Mixed)	0 and 1	1 ($S_{\text{particle}}=0$) / 0 ($S_{\text{particle}}=1$)
Cluster Transfer	$(^6\text{Li}, d)$	Transfer of α -cluster	0	0
Charge Exchange	$(p, n), (n, p)$	Fermi / Gamow-Teller	0 and 1	1
	$(^3\text{He}, t), (t, ^3\text{He})$	Fermi / Gamow-Teller	0 and 1	1
	$(^6\text{Li}, ^6\text{He})$	Pure Gamow-Teller ($1^+ \rightarrow 0^+ + \alpha$)	1	1